# ECS7032P - Assignment 1 - Assessment Rubric

| Assessment Area | Assessment Item | Determined by: | 70%-100% | 50%-69% | < 50% |
|---|---|---|---|---|---|
| 1) Abstract | Brief motivation of the work | Report (Abstract) | Clear motivation, correct and succint. | The motivation is stated, although it is incomplete, has minor errors or is not properly summarized. | The work is not motivated in the abstract, or is not correct. |
| | Brief summary of the work | Report (Abstract) | Work described appropriately, including all key elements and at the appropriate length. | Work described to an acceptable degree, missing some elements. | The work is not summarized in the abstract, or is not correct. |
| | Outline of the findings | Report (Abstract) | The abstract succintly describes the main experiments done, as well as the most interesting findings. | The abstract indicates experiments were made and some results are mentioned, but they are not described at the appropriate length, or the focus is not set in the most interesting results. | The abstract makes no mention of findings or experiments being made. |
| 2) Problem Definition | Motivation of the work | Report (Introduction) | Clear motivation of the work done, stating its importance and citing relevant works in the literature that study similar concepts. | There is some description of the motivation of the work, indicating why is it interesting to study the topic of this project. Incomplete citations to similar works in the literature. | The motivation of the work is not described in the introduction. |
| | Clarity of the problem definition | Report (Introduction) | Problem described at the right length, indicating clearly what are the main research challenges of this project. | Decent description of the problem that was addressed, indicating some of the research challenges that are present. | The problem is not defined properly, missing the key challenges that make this work interesting. |
| | Clarity of the contributions of the report | Report (Introduction) | Clear description of the contributions of this work, highlighting its importance to the advancement of knowledge. | There is some description of the contributions of this work, but it is either incomplete, missing the most important contributions, or some contributions listed are not valid research contributions. | It is not clear, by reading the introduction, of what has been done that constitutes and advancement in knowledge. |
| | Critical Analysis of the literature | Report (Background) | Deep analysis of the literature from a critical point of view, highlighting the gaps in the literature that can be addressed by further studies. | Sufficient literature is analysed in the background section, although there is a lack of a deep critical analysis of the state of the art. Gaps in research are only briefly mentioned, or not mentioned at all. | Insufficient literature is analysed in terms of methods, environments or approaches that are relevant to the current study. |
| | Description of the game (Sushi Go!) | Report (Background) | Clear description of the game and its rules, highlighting the main characteristics that make it interesting for AI research. | Sufficient description of the environment used for the assignment, including the main rules of the game. | Insufficient or no description of the environment used for experimentation. |
| 3) Approach | Understanding and application of the concepts relevant to Statistical Planning | Report (Methods) and Code | Algorithms and/or improvements are applied and implemented correctly, going beyond what has been seen in the lectures. | Algorithms and/or improvements are applied and implemented correctly, following the principles seen in lectures. Minor errors or omissions in complex code implementations may exist. | The application of algorithms and/or improvements to these is incorrect or has important flaws in either ideation or implementation. |
| | Originality in tackling the problem | Report (Methods) | The methods employed combine elements seen in the lectures with other aspects that go beyond what's seen in the course or in published literature. | The methods employed combine elements seen in the lectures or research papers sensibly. | The methods employed are a replica of other algorithms present in the framework, without any innovation or modifications. |
| | Difficulty level and ambition of the presented approach | Report (Methods) | Implementation or improvement of advanced tree search or evolutionary agents, or tuning of agent parameters with a substantial development. | Improvements on simple agents (rule based, OSLA) or tuning parameters of existing agents without any substantial development. | Implementation of agents that do not go beyond the complexity of simple rule based, one step look ahead or random agents. |
| 4) Experimentation | Adequacy and rigour of the experimental work | Report (Experimental Study) | The agent developed is appropriately compared against more than one agent in the framework, plus against a base version of the agent that has been implemented (e.g. without improvements made to that base).<br>Thorough ablation studies comparing the contribution of different elements of the approach. | The agent developed is appropriately compared against other sample agents in the framework and/or a base version of the agent that has been implemented (e.g. without improvements made to that base). Some simple ablation studies may be present at the higher level in this categeory. | The type of experimentation and the number of games played between the agents is incorrect or insufficient to draw any conclusions.<br><br>[We deem running less than 500 games per pair of agents as insufficient for this assignment.] |
| | Quality of the documentation of the experimental work | Report (Experimental Study) | The experimental process can be replicated externally with the information given in the report. | There is an attempt to include in the report documentation (games played, configuration of these games, parameters of the agents) to allow replication of the experimental process, with only a few omissions or details. | The report does not include sufficient documentation, in terms of games played, configuration of these games or parameters of the agents, to replicate the experimental process externally. |
| | Critical analysis of the results | Report (Experimental Study) | The most important aspects of the experimental process are properly described, with hypothesis or explanations that indicate the nature of the observed results.<br>Good error analysis. | There is an attempt to highlight the most important aspects found in the experimental process, and explanations are made to try to explain the results observed. This explanations do not go beyond the obvious or contain minor mistakes.<br>Some simple analysis of errors. | Results are presented raw, either in tables or in textual formal, without any critical analysis that highlights the most important aspects, or attempts to explain / hypothesize the reasons behind the results obtained.<br>Absence of any error analysis. |
| 5) Conclusions | Appropriate summary and conclusions of the work | Report (Conclusions) | The conclusions appropriately summarize the contents of the report, highlighting the most important aspects of the work. | The conclusions summarize the work done in the project, but there are certain omissions or misrepresentations of what is mentioned in the report. | The conclusions do not properly summarize the work done in the project. |
| | Presentation of possibilities for further work | Report (Conclusions) | Future work is seriously included in the report, with sensible future steps and reasoning on why that was selected as further work. | Some ideas for future work are presented in the report, aligned with the work done in this project. | Ideas for future work are not presented in the report, or are not appropriate for this study. |
| 6) General | Clarity of ideas and their expression in the document | Report (whole document) | Very good write-up with a logical structure, good flow, technically precise and concise style. | Adequate write-up, lacking clarity or detail in places, or containing irrelevant material. Good use of technical language. It is easy to understand the core ideas. | The write-up is unclear, disorganized and/or written badly. It is very difficult to understand core ideas. |
| | Quality of figures, tables and legends | Report (whole document) | Figures demonstrate conceptual thinking; graphs are fully annotated, are easy to read and interpret, provide insight, and fully support claims and conclusions; figure legends are concise and informative. | Figures and figure legends are of good quality and are helpful for understanding the project. | Figures and figure legends are of insufficient quality. |
| | Correct referencing | Report (whole document) | Referencing is employed throughout following the discipline standard. There are no omissions of relevant literature in the scope of the project. | Appropriate referencing supports claims well and uses consistent format throughout. Some key references may be missed, or irrelevant ones included. | There is no referencing, or it is inappropriate to the work, or done incorrectly. |